

Thermodynamic Modeling: Ethylenediaminetetraacetic acid tetrasodium salt (Na₄EDTA)

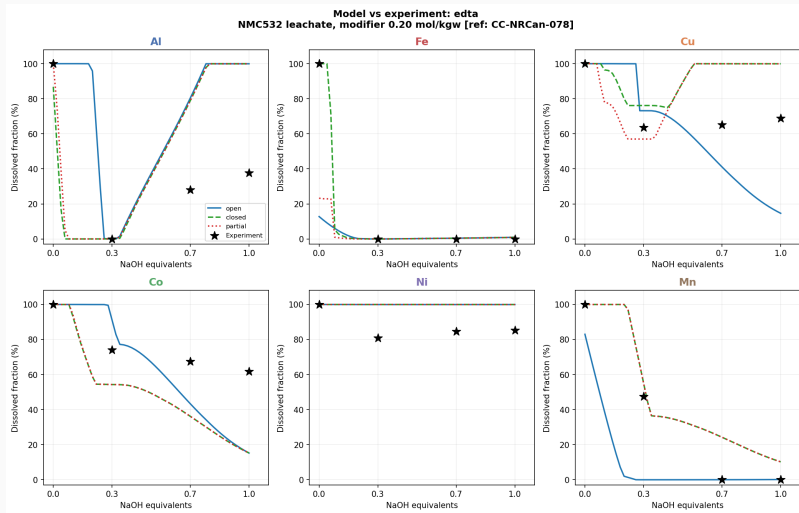
NMC532 Leachate, Modifier 0.20 mol/kgw

MDPSA Pipeline

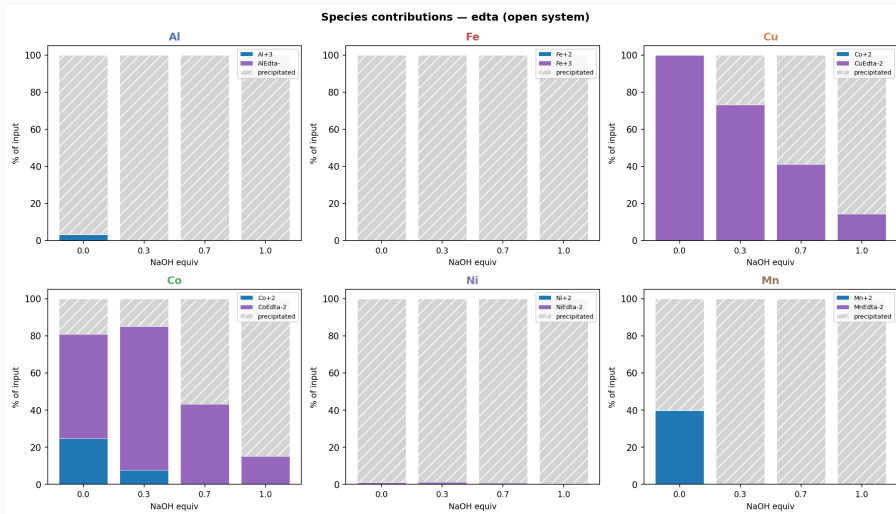
2026-07-22

NRCan CC-NRCan-078

Model vs. Experiment



Species Contributions (Open System)



Selectivity Scorecard

Metal	Role	@ 0.3 eq	@ 1.0 eq	RMSE (pp)	Verdict
Cu	Target	73%	15%	30.0	Partial
Co	Target	91%	15%	27.6	Partial
Ni	Target	100%	100%	14.3	Protected
Mn	Target	0%	0%	25.2	Lost
Al	Impurity	0%	100%	40.7	Partial
Fe	Impurity	0%	1%	43.6	Removed

RMSE: root-mean-square error in percentage points vs. experiment (CC-NRCan-078).

Protected/Removed: model matches experiment at all comparison equiv points.

Thermodynamic Parameters: Ethylenediaminetetraacetic acid tetrasodium salt (Na₄EDTA)

Species	log K
CuEdta-2	20.50
CoEdta-2	16.31
NiEdta-2	18.62
MnEdta-2	13.87
AlEdta-	16.30

- Database: sit.dat (SIT activity model)
- Modifier dose: 0.20 mol/kgw
- Temperature: 25°C, I = 0 (NIST reference)
- Equilibrium phases: Gibbsite, Ferrihydrite(am), Cu(OH)₂(s), Co(OH)₂(s), Ni(OH)₂(s), MnOOH
- Fe complexation suppressed (kinetic)
- Scenarios: open / closed / partial CO₂

Source: NIST Standard Reference Database 46 Version 8.0. Values adjusted for ionic-strength effects at leachate conditions.